



**2016 Year 12
HSC Assessment Task 1
December**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 50 minutes
- Write using non erasable black or blue pen
- Board-approved calculators may be used
- Show all necessary working
- Marks may be deducted for careless or badly arranged work

Total marks – 33

Questions 1-3 (pages 2-4)

Answer each question on the appropriate pages of your answer booklet.

Question 1 (11 marks) - Use the Question 1 section of the writing booklet.		Marks
a)	Find the value of k if $(x - 2)$ is a factor of $P(x) = x^3 - kx + 6$	1
b)	If α, β and γ are the roots of the equation $2x^3 - 3x - 4 = 0$ Find the value of (i) $\alpha + \beta + \gamma$ (ii) $\alpha\beta + \alpha\gamma + \beta\gamma$ (iii) $\alpha\beta\gamma$ (iv) $\alpha^2 + \beta^2 + \gamma^2$	1 1 1 2
c)	Given the polynomials $f(x) = 2x^4 - 10x^3 + 12x^2 + 2x - 3$ and $g(x) = x^2 - 2x + 1$. Find expressions for $A(x)$ and $R(x)$ if $f(x) = g(x)A(x) + R(x)$.	2
d)	For all positive integers of n, use mathematical induction to prove $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \cdots + \frac{1}{(n+1)(n+2)} = \frac{n}{2(n+2)}$	3

Question 2 (10 marks) - Use the Question 2 section of the writing booklet.	Marks
a) When the polynomial $P(x)$ is divided by $x^2 - 4$ the remainder is $3x - 1$. What is the remainder when $P(x)$ is divided by $x + 2$	2
b) Write an equation for the polynomial function below. The figure shows a graph of a polynomial function on a Cartesian coordinate system. The x-axis and y-axis are shown. The curve has x-intercepts at -1 and 3. It passes through the point (2, 1). The text "Not to scale" is written in the upper right area of the graph.	2
c) One of the roots of the equation $x^3 + kx^2 + 1 = 0$ is equal to the sum of the other two roots. (i) Show that $x = -\frac{k}{2}$ is a root of the equation. (ii) Find the value of k.	2 2
d) Show $(x - 1)(x - n)$ is a factor of $P(x) = x^{2n} - x^n + n^n - (xn)^n$ where n is a positive integer.	2

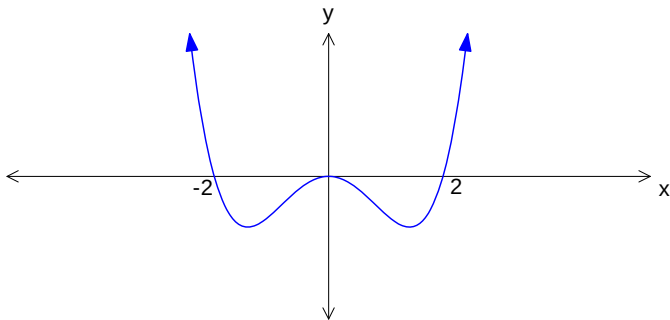
Question 3 (12 marks) - Use the Question 3 section of the writing booklet.		Marks
a)	An odd polynomial $P(x)$ of smallest degree has double root at $x = 2$ and the gradient of the tangent at $x = 1$ is -6 . Find an equation for $P(x)$.	3
b)	(i) Sketch the polynomial $P(x) = x^4 - 4x^2$	2
	(ii) Show that $P(x)$ has stationary points at $(0,0)$, $(-\sqrt{2}, -4)$ and $(\sqrt{2}, -4)$.	2
	(iii) Hence, or otherwise, find all the values of k such that $P(x) + k = 0$ will have exactly two solutions.	2
c)	Use mathematical induction for prove $n^3 - n$ is divisible by 6 for all positive integers of n .	3
End of the Exam		

BAULKHAM HILLS HIGH SCHOOL - EXTENSION 1
2016 HSC Task 1 SOLUTIONS

Solution		Mks	Comments
QUESTION 1			
1a.	$2^3 - k(2) + 6 = 0$ $k = 7$	1	1 mark • Correct substitution.
1b.(i)	$\alpha + \beta + \gamma = 0$	1	1 mark • Correct answer.
1b.(ii)	$\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{3}{2}$	1	1 mark • Correct answer.
1b.(iii)	$\alpha\beta\gamma = 2$	1	1 mark • Correct answer.
1b.(iv)	$(\alpha + \beta + \gamma)^2 = (\alpha + \beta)^2 + 2\gamma(\alpha + \beta) + \gamma^2$ $(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + 2\alpha\beta + 2\alpha\gamma + 2\beta\gamma + \gamma^2$ $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ $= (0)^2 - 2\left(-\frac{3}{2}\right)$ $= 3$	2	2 marks • Correct solution 1 mark • Significant progress to finding an expression for $\alpha^2 + \beta^2 + \gamma^2$
1 c.	$2x^4 - 10x^3 + 12x^2 + 2x - 3 = (x^2 - 2x + 1)(2x^2 - 6x - 2) + 4x - 1$ $A(x) = 2x^2 - 6x - 2$ $R(x) = 4x - 1$	2	2 marks • Correct expressions 1 mark • Significant progress towards solution.
1d	Test for $n = 1$ $LHS = \frac{1}{(1+1)(1+2)} = \frac{1}{6}$ $RHS = \frac{1}{2(1+2)} = \frac{1}{6}$ $\therefore$ true for $n = 1$ Assume true for $n = k$ $ie. \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \dots + \frac{1}{(k+1)(k+2)} = \frac{k}{2(k+2)}$ Prove true for $n = k + 1$ Aim: $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{(k+1)(k+2)} + \frac{1}{(k+2)(k+3)} = \frac{k+1}{2(k+3)}$ $LHS = \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{(k+1)(k+2)} + \frac{1}{(k+2)(k+3)}$ $= \frac{k}{2(k+2)} + \frac{1}{(k+2)(k+3)} \times \frac{2}{2} \quad \text{if the assumption is true}$ $= \frac{1}{2(k+2)} \left[\frac{k(k+3)}{(k+3)} + \frac{2}{(k+3)} \right]$ $= \frac{1}{2(k+2)} \left[\frac{k(k+3) + 2}{(k+3)} \right]$ $= \frac{1}{2(k+2)} \left[\frac{k^2 + 3k + 2}{(k+3)} \right]$ $= \frac{1}{2(k+2)} \left[\frac{(k+2)(k+1)}{(k+3)} \right]$ $= \frac{k+1}{2(k+3)}$ If it is true for $n = k$, then it is also true for $n = k + 1$. By Induction, since it is true for $n = 1$, then it is also true for $n = 2, 3, 4, \dots$ all positive integer of n	3	There are 4 key parts of the induction; - Proving the result true for $n = 1$ - Clearly stating the assumption and what is to be proven - Using the assumption in the proof - Correctly proving the required statement 3 marks • Successfully does all of the 4 key parts 2 marks • Successfully does 3 of the 4 key parts 1 mark • Successfully does 2 of the 4 key parts

Solution		Mks	Comments
QUESTION 2			
2a.	$P(x) = (x^2 - 4)Q(x) + 3x - 1$ $P(x) = (x + 2)(x - 2)Q(x) + 3x - 1$ $P(-2) = (-2 + 2)(-2 - 2)Q(x) + 3(-2) - 1$ $P(-2) = -7$ $\text{ie. } R(-2) = 3(-2) - 1 = -7$	2	2 mark • Correct solution. 1 mark • Significant attempts to use the remainder theorem
2b.	$P(x) = a(x + 1)^2(x - 3)^3$ At (2,1) $1 = a \times 2^2(2 - 3)^3$ $a = -\frac{1}{9}$ $P(x) = -\frac{1}{9}x^2(x - 3)^3$	2	2 marks • Correct solution 1 mark • Correctly identifies $P(x) = (x + 1)^2(x - 3)^3$
2c (i)	Let the roots be α, β and $\alpha + \beta$ Sum of roots: $\alpha + \beta + (\alpha + \beta) = -k$ $2(\alpha + \beta) = -k$ $(\alpha + \beta) = -\frac{k}{2}$ Since $\alpha + \beta$ is a root, then $-\frac{k}{2}$ is a root	2	2 marks • Correct solution 1 mark • Correctly identifies sum of roots = $-a$ • Identify the third root as $\alpha + \beta$
2c.(ii)	Since $-\frac{k}{2}$ is a root, then $\left(-\frac{k}{2}\right)^3 + k\left(-\frac{k}{2}\right)^2 + 1 = 0$ $-\frac{k^3}{8} + \frac{k^3}{4} = -1$ $k^3 = -8$ $k = -2$	2	2 marks • Correct solution 1 mark • Correctly identifies $-\frac{k}{2}$ is a root
2d	$P(1) = 1^{2n} - 1^n + n^n - (1n)^n$ $= 1 - 1 + n^n - n^n$ $= 0$ $\therefore (x - 1) \text{ is a factor}$ $P(n) = n^{2n} - n^n + n^n + (n \times n)^n$ $= n^{2n} - n^n + n^n + (n^2)^n$ $= 0$ $\therefore (x - n) \text{ is a factor}$ $\therefore (x - 1)(x - n) \text{ is a factor of } P(x)$	2	2 marks • Correct solution 1 mark • Correctly proves one factor.

QUESTION 3

Solution		Mks	Comments
3a.	If $P(x)$ is odd, it must pass through the origin. Since $P(x)$ has a double root at $x = 2$, then $(x - 2)^2$ is a factor. Since $P(x)$ is odd, $(x + 2)^2$ is also a factor. $\therefore P(x) = ax(x - 2)^2(x + 2)^2$ If the gradient of the tangent -6 at $x = 1$ then $P'(1) = -6$ $P(x) = ax(x^2 - 4)^2$ $P(x) = ax(x^4 - 8x^2 + 16)$ $P(x) = ax^5 - 8ax^3 + 16ax$ $P'(x) = 5ax^4 - 24ax^2 + 16a$ $P'(1) = 5a - 24a + 16a$ $-6 = -3a$ $a = 2$ $\therefore P(x) = 2x(x - 2)^2(x + 2)^2$	3	1. -Recognises it passes the origin. ie. a factor of x. -Recognises $(x - 2)^2(x + 2)^2$ is a factor. -Recognises a factor of a. 2. Finds the derivative 3. Uses derivative to solve for a 3 marks • Correct Solution. 2 mark • Successfully does 2 parts. 1 mark • Successfully does 1 parts.
3b.(i)		2	2 marks • Correct answer. 1 mark • Correct graph without any intercepts.
3b (ii)	$P'(x) = 4x^3 - 8x$ Stationary point when $P'(x) = 0$ $0 = 4x(x^2 - 2)$ $x = 0, \sqrt{2}, -\sqrt{2}$ $P(0) = 0$ $P(\sqrt{2}) = \sqrt{2}^4 - 4\sqrt{2}^2$ $= -4$ $P(-\sqrt{2}) = (-\sqrt{2})^4 - 4(-\sqrt{2})^2$ $= -4$	2	2 marks • Correct solution 1 mark • Finds all stationary points without y-values
3b.(iii)	$k < 0$ or $k = 4$	2	2 marks • Correct solution 1 mark • One region correct.

Solution		Mks	Comments
3c	<u>Test for $n = 1$</u> $LHS = 1^3 - 1$ $= 0$ Since 0 is divisible by 6 $\therefore$ true for $n = 1$ <u>Assume true for $n = k$</u> ie. $k^3 - k = 6P$ where P is a positive integer Prove true for $n = k + 1$ Aim: $(k + 1)^3 - (k + 1) = 6Q$ where P is a positive integer $LHS = (k + 1)^3 - (k + 1)$ $= k^3 + 3k^2 + 3k + 1 - k - 1$ $= k^3 + 3k^2 + 2(k^3 - 6P)$ if the assumption is true $= 3k^3 + 3k^2 - 12P$ $= 3k^2(k + 1) - 12P$ $\therefore k(k + 1) = 2M$ since k and $(k + 1)$ are consecutive number, one of them must be even. $LHS = 3k \times 2M - 12P$ $= 6(km - 2P)$ where $(km - 2P)$ is an integer $\therefore (k + 1)^3 - (k + 1)$ is divisible by 6. If it is true for $n = k$, then it is also true for $n = k + 1$. By Induction, since it is true for $n = 1$, then it is also true for $n = 2, 3, 4 \dots$ all positive integer of n	3	There are 4 key parts of the induction; 1. Proving the result true for $n = 1$ 2. Clearly stating the assumption and what is to be proven 3. Using the assumption in the proof 4. Correctly proving the required statement (must explain why $k(k + 1) = 2M$) 3 marks • Successfully does all of the 4 key parts 2 marks • Successfully does 3 of the 4 key parts 1 marks • Successfully does 2 of the 4 key parts Note: Without using the assumption, it is not the process of mathematical induction